🚚 SUPPLY CHAIN ANALYTICS

Natural Language Supply ChainAssistant

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Project Overview

- Developed an AI-powered supply chain analytics assistant
- Converts natural language business questions into Spark SQL queries
- Executes queries on Databricks warehouse and visualizes results
- Provides interactive chatbot interface using Streamlit

Problem Statement

- Manual data analysis is time-consuming
- Business users need technical SQL knowledge to query data
- Supply chain performance insights are difficult to generate quickly
- Lack of real-time interactive analytics dashboard

Architecture

User Question → Streamlit UI

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Mistral LLM SQL Generator

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Databricks SQL Warehouse

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Supply Chain Silver Table

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Visualization Output

Key Features

- Natural Language Query Interface
- Automated SQL Query Generation
- Supply Chain Analytics Chatbot
- Secure Databricks SQL Execution
- Chat Memory Support
- KPI Visualization Charts
- Delivery delay and performance analysis

Business Impact

- Reduces manual reporting effort
- Improves decision making speed
- Provides real-time supply chain insights
- Enhances delivery performance monitoring
- Supports data-driven operational strategy

Future Enhancements

- Predictive inventory shortage forecasting
- Advanced machine learning risk scoring
- Multi-table semantic search capability
- Deployment as enterprise analytics assistant
- Voice-based query interface
- Alert automation for late shipments

Conclusion

- AI-driven analytics improves supply chain visibility
- Natural language interface simplifies data access
- Scalable Lakehouse architecture supports large datasets
- Helps business users make faster operational decisions

Thank You